

Analyzed File	Aeolus Task 2	Version	Autodesk Fusion (2704.1.53)
Creation Date	2026-09-11, 23:34:03	Author	NEEL

Report Properties

Report Properties	
Title	Studies
Author	NEEL

Simulation Model 2

Study 1 - Static Stress

Study Properties

Study Properties	
Study Type	Static Stress
Last Modification Date	2026-09-11, 23:19:11

Study Setup

Settings		
General		
	Contact Tolerance	0.10 mm
	Remove Rigid Body Modes	No
Mesh		
	Average Element Size (% of model size) - Solids	10
	Scale Mesh Size Per Part	No
	Average Element Size (absolute value)	-
	Element Order	Parabolic
	Create Curved Mesh Elements	No
	Max. Turn Angle on Curves (Deg.)	60
	Max. Adjacent Mesh Size Ratio	1.5
	Max. Aspect Ratio	10
	Minimum Element Size (% of average size)	20
Adaptive Mesh Refinement		
	Number of Refinement Steps	0
	Results Convergence Tolerance (%)	20
	Portion of Elements to Refine (%)	10
	Results for Baseline Accuracy	von Mises Stress

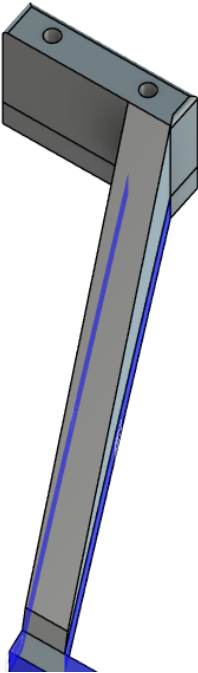
Materials		
Steel		
Components		
	Component	Strength
	Body1	Yield Strength
	Component1:1	Yield Strength
Properties		
	Density	7.850E-06 kg / mm^3
	Young's Modulus	210000.00 MPa
	Poisson's Ratio	0.30
	Yield Strength	207.00 MPa
	Ultimate Tensile Strength	345.00 MPa
	Thermal Conductivity	0.056 W / (mm K)
	Thermal Expansion Coefficient	1.200E-05 / K
	Specific Heat	480.00 J / (kg K)

Contacts

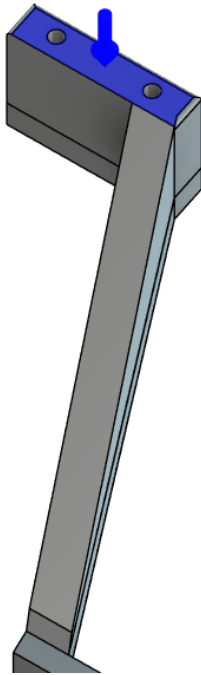
Contact Type	Contact Set	Penetration Type	Bodies	Entities
Bonded				
	Bonded1	Symmetric	[Body1 Simulation Model 2:1 Body1 Component1:1]	[Face 13 Face 11]

Mesh	
Type	Solids
Nodes	6549
Elements	3502

Load Case1

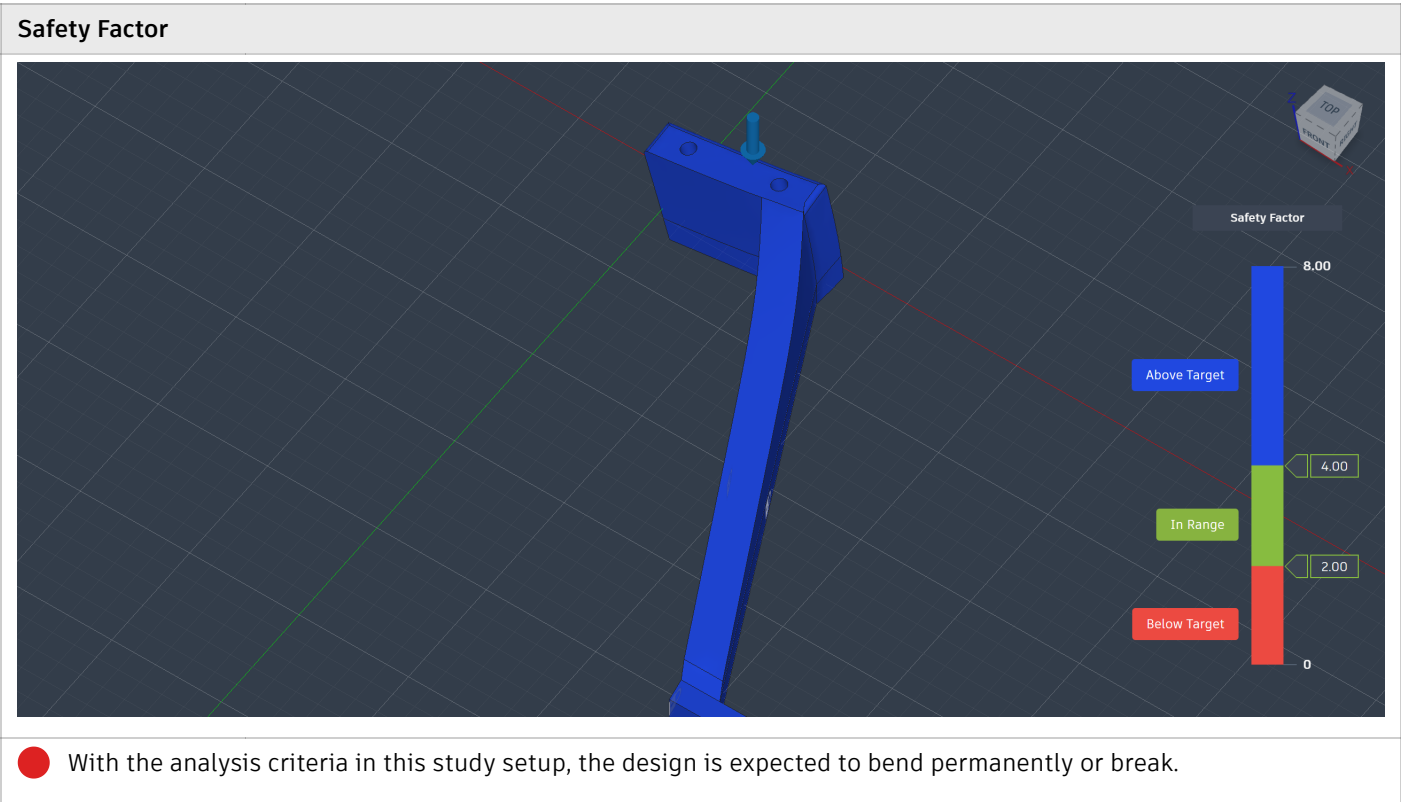
Constraints	
Fixed1	
	
Type	Fixed
Ux	Fixed
Uy	Fixed
Uz	Fixed

Loads	
Force1	



Type	Force
Magnitude	40.00 N
X Value	0.00 N
Y Value	0.00 N
Z Value	-40.00 N
Force Per Entity	No

Load Case1 - Guided Results

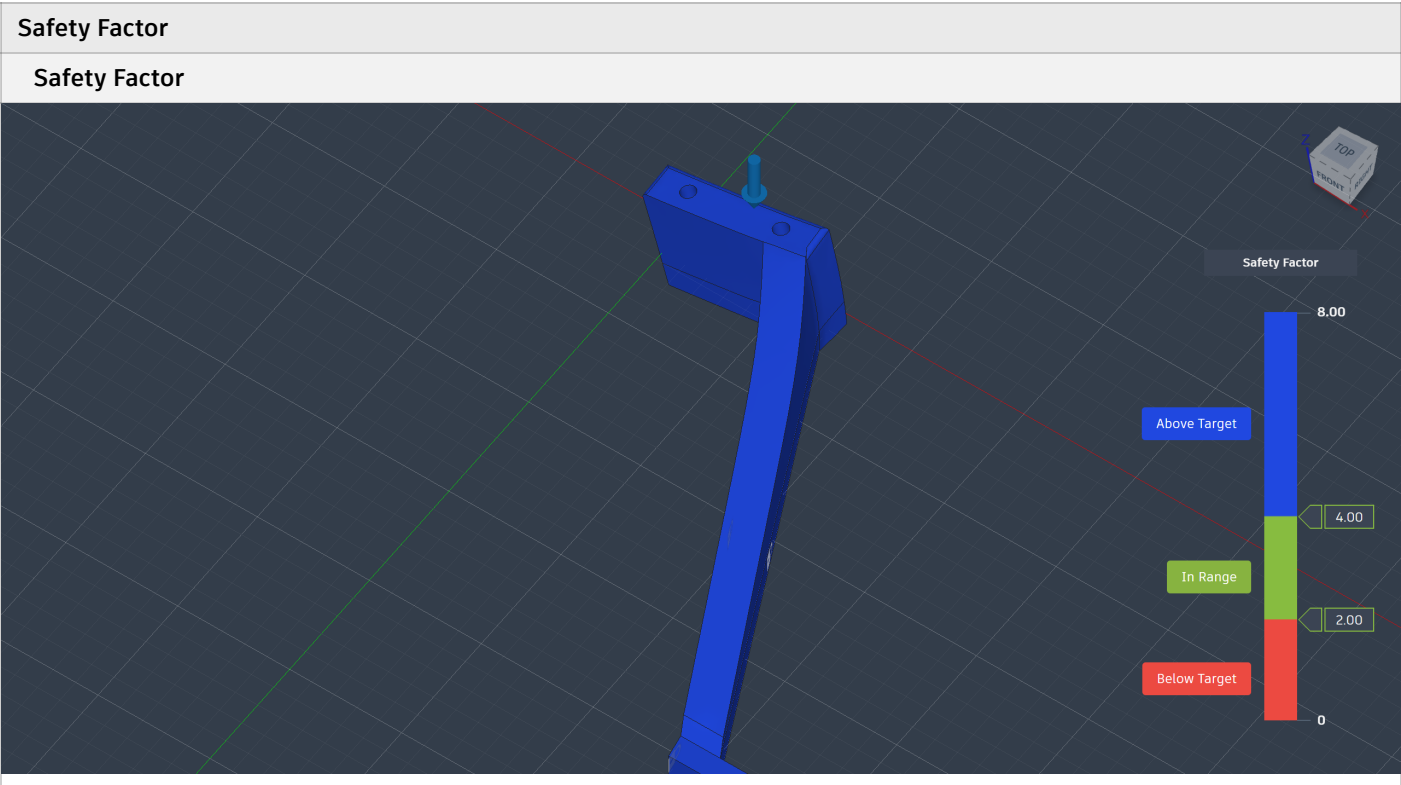


Next Steps	Below Safety Factor Target
	Validate your study setup • Check the loads are not set too high • Check the mesh refinement and consider increasing it in areas of high stress • Confirm your Safety Factor requirement • If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight
	Check other results • Examine the Below target Result visualization to see where the weakest areas are • Check the stress result to see if, and where, the material might yield • Check the displacement result to see how far from its original alignment the model has moved • Look at other results, such as reaction forces to check your study set up, or strain to check your material performance
	Strengthen weak areas • Add more material to weak areas • Select a material with a higher yield strength • Stiffen the design • Modify the shape with a Generative Design study
	Generate alternative designs • Add more material to weak areas • Select a material with a higher yield strength • Stiffen the design • Modify the shape with a Generative Design study
	Above Safety Factor Limit
	Validate your study setup • Check the loads are not set too high • Check the mesh refinement and consider increasing it in areas of high stress • Confirm your Safety Factor requirement • If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight
	Check other results • Examine the Below target Result visualization to see where the weakest areas are • Check the stress result to see if, and where, the material might yield • Check the displacement result to see how far from its original alignment the model has moved • Look at other results, such as reaction forces to check your study set up, or strain to check your material performance
	Lightweight the design • Set a Safety Factor upper limit to find where material can be removed • Remove material from areas that exceed the Safety Factor upper limit
	Generate alternative designs • In the Generative Design workspace, create design modifications • Remove unnecessary material using a Shape Optimization study

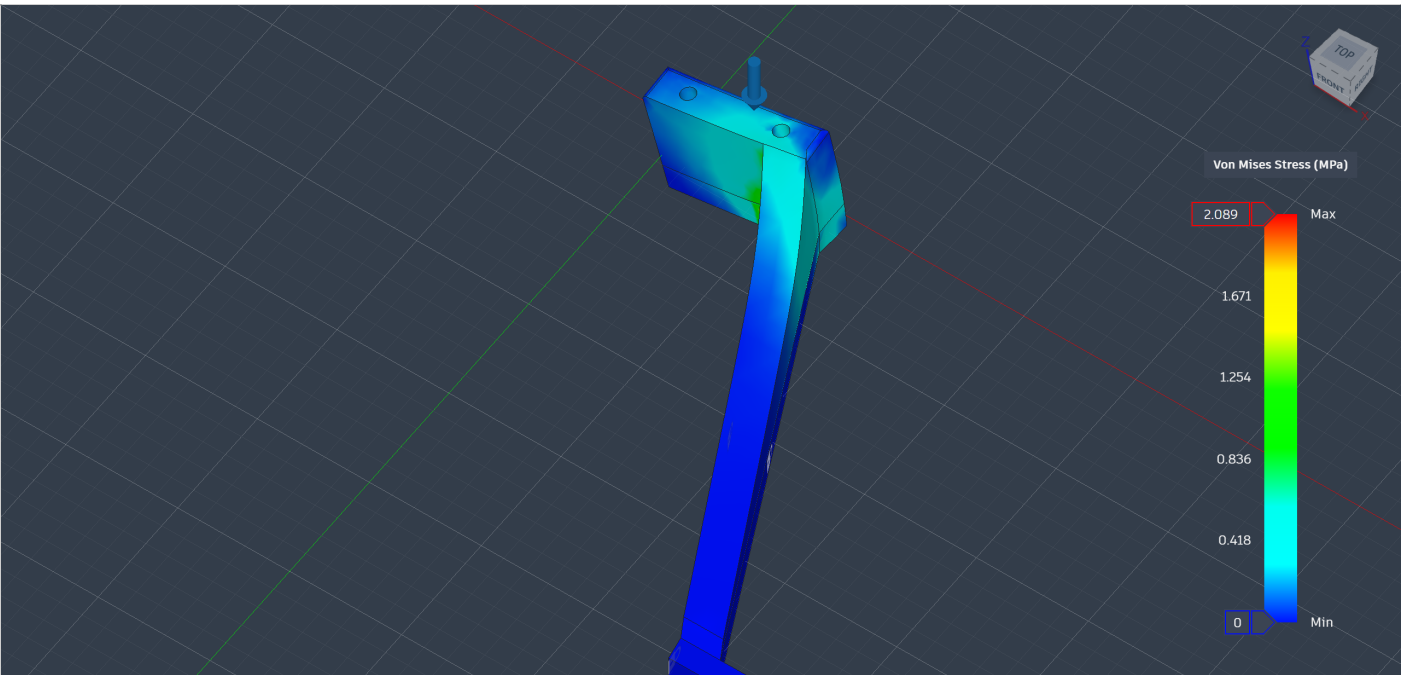
Load Case1 - Results

Result Summary			
Result type	Sub-result	Minimum	Maximum
Safety Factor			
	Safety Factor	15.00	1.545E+11
Stress			
	Von Mises	0 MPa	2.089 MPa
	1st Principal	-0.601 MPa	0.995 MPa
	3rd Principal	-2.736 MPa	0.193 MPa
	Normal XX	-2.33 MPa	0.628 MPa
	Normal YY	-0.743 MPa	0.571 MPa
	Normal ZZ	-2.659 MPa	0.665 MPa
	Shear XY	-0.439 MPa	0.673 MPa
	Shear YZ	-0.744 MPa	0.24 MPa
	Shear XZ	-0.327 MPa	0.88 MPa
Displacement			
	Total	0 mm	3.725E-04 mm
	X	-1.855E-04 mm	6.113E-05 mm
	Y	-7.30E-05 mm	1.427E-05 mm
	Z	-3.221E-04 mm	2.768E-05 mm
Reaction Force			
	Total	0 N	28.272 N
	X	-8.528 N	3.998 N
	Y	-1.337 N	4.042 N
	Z	-10.124 N	27.982 N
Strain			
	Equivalent	0	1.48E-05
	1st Principal	0	9.43E-06
	3rd Principal	-1.532E-05	0
	Normal XX	-9.704E-06	2.62E-06
	Normal YY	-3.003E-06	2.577E-06
	Normal ZZ	-1.046E-05	2.591E-06
	Shear XY	-5.433E-06	8.337E-06
	Shear YZ	-9.211E-06	2.974E-06

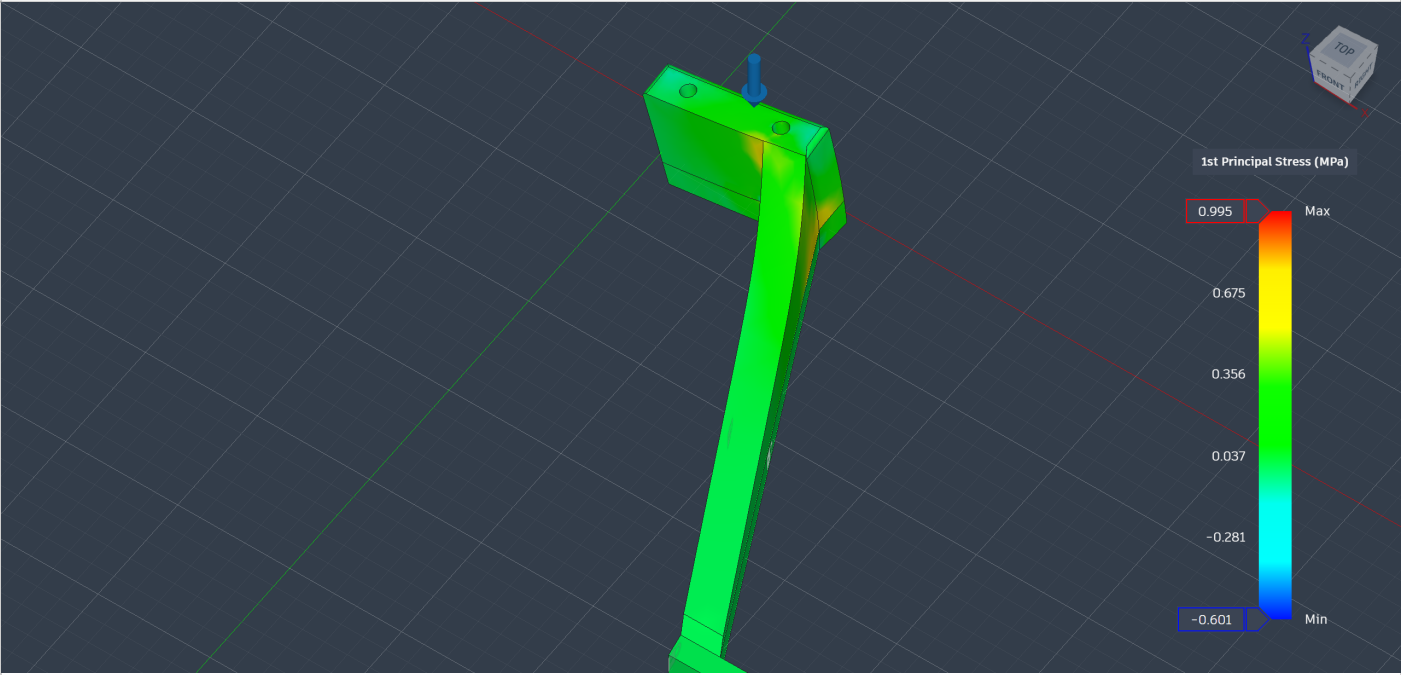
	Shear XZ	-4.048E-06	1.09E-05
Contact Pressure			
	Total	0 MPa	2.689 MPa
	X	-0.464 MPa	0.416 MPa
	Y	-0.446 MPa	0.46 MPa
	Z	-0.665 MPa	2.659 MPa
Contact Force			
	Total	0 N	22.807 N
	X	-3.889 N	3.889 N
	Y	-3.875 N	3.872 N
	Z	-20.995 N	22.457 N



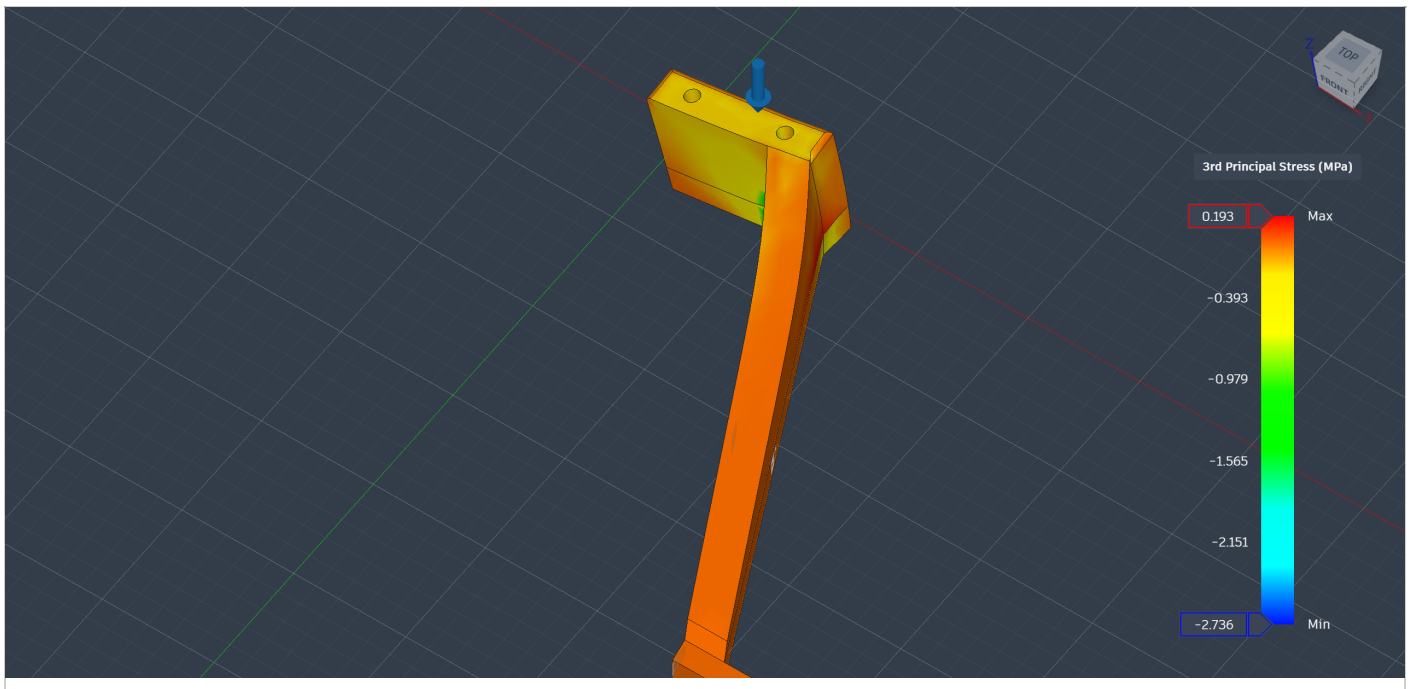
Stress
Von Mises



1st Principal



3rd Principal



Displacement

Total

